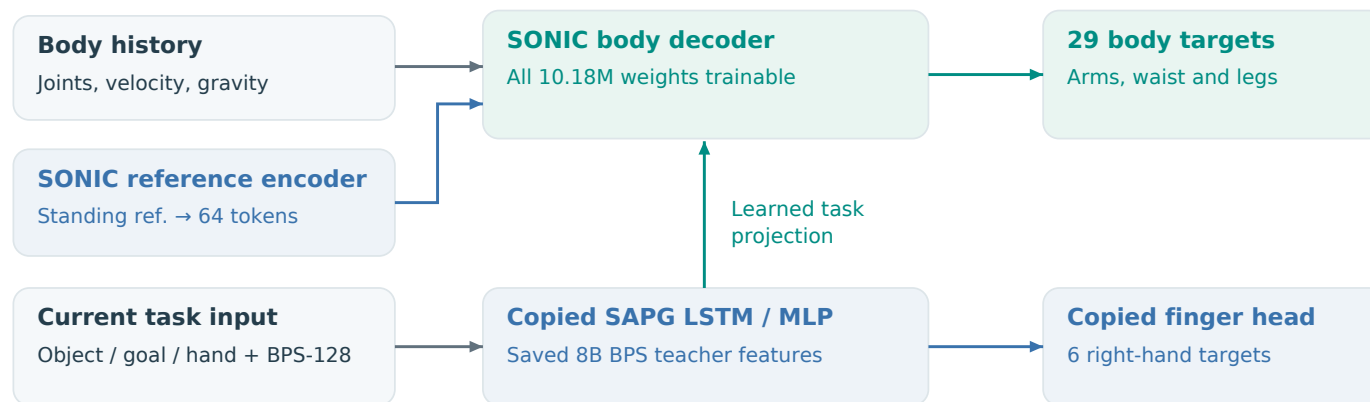


Teaching SONIC to manipulate while standing

Reuse a trained SAPG skill in one full-body G1 + Revo2 controller. One RTX 6000 Ada, 48 GB.

IMPLEMENTED: INITIAL DISTILLATION | NOT YET STARTED: FULL-BODY SAPG RL

01 One student, two pretrained sources



35 actions • no separate SAPG arm override • copied task/finger weights frozen only during bootstrap

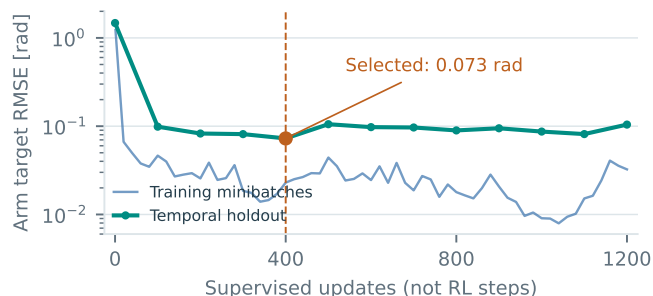
02 Full-physics setup



Initial CAD scene—not a successful grasp.

03 Measured skill transfer

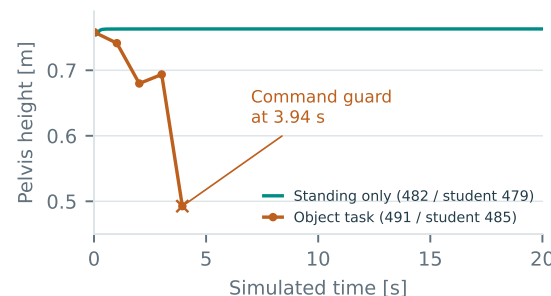
4 × 30 s clips; brush, eraser, hammer, spatula



Temporal split + gap; synthetic standing-body bootstrap.
1,359 scored holdout samples. Error ≠ grasp success.

04 Closed-loop evidence

Standing passes; manipulation does not yet.



Min. over 4 envs; different settings—not an ablation.
20 s standing passed; object task stopped at 3.94 s.

05 Proposed next stages

- 1 Validate loaded physics**
Resolve contact instability; add goal/reset logic.
- 2 Fine-tune the same student with SAPG**
35-action actor/critic + teacher loss; learn balance.
- 3 Integrate and size the complete workload**
70 Hz tactile; benchmark rollouts + optimizer on 48 GB.
- 4 Expand after right-hand validation**
Disturbances and payloads, then two hands: 41 actions.

Why this route: SAPG provides manipulation skill; SONIC provides the body-control initialization. Imitation transfers arm targets and finger commands, but online full-body RL must learn balance and load transfer. **Both hands, feet, gravity and self-contact stay in physics; no fabrics/PCA.**

Evidence: 1,200 supervised updates; 20 targeted tests passed. No validated standing-manipulation success rate yet.